CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2012

EC207 Control Systems

Date: 16.05.2012, Wednesday | Time: 01:30 p.m. To 04:30 p.m. | Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use graph papers.

SECTION – I

Q - 1 The open loop transfer function of a unity feedback control system is $\frac{1}{s^2(1+s)(1+2s)}$. Using Nyquist Criterion, Determine whether the closed loop system is stable or not. [07]

Q – 2(a) A feedback system has $G(s)H(s)=\frac{100(s+4)}{s(s+0.5)(s+10)}$ [07]

Draw the bode-plot and comment on stability.

(b) For the given characteristics equation, find the no. of roots in Right Half of S-plane. [07]

$2s^6+4s^5+s^4-32s^3+51s^2+3s+5$

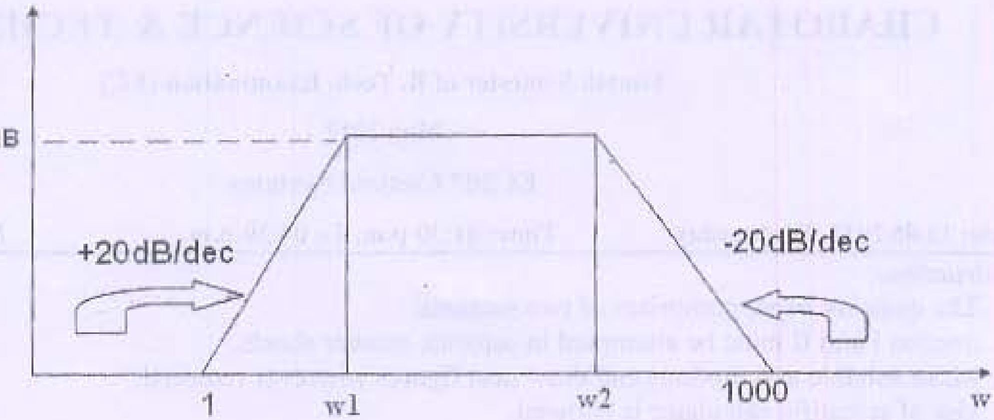
OR

Q - 2(a) List out the necessary conditions for stability [07]

(b) Find range of values of 'K' so that the system with following characteristics equation will be Stable, Unstable and Critically stable. $s(s^2+s+1)(s+4)+K=0$ [07]

Q - 3 [07]

(a)



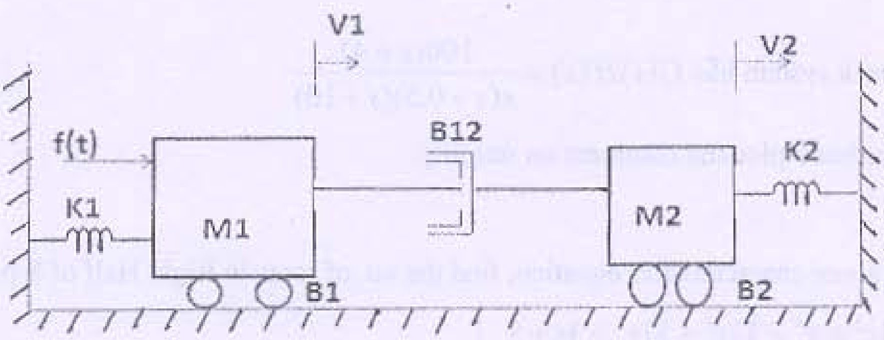
From the given bode plot find out the transfer function.

(b) Draw the polar plot for the transfer function and comment on stability. [07]

$$G(s)H(s) = \frac{10(s+1)}{(s+10)}$$

OR

Q -3(a) For the mechanical system as in figure. Draw the F $\rightarrow$ V and F$\rightarrow$ I analogous Electricalcircuits. [07]

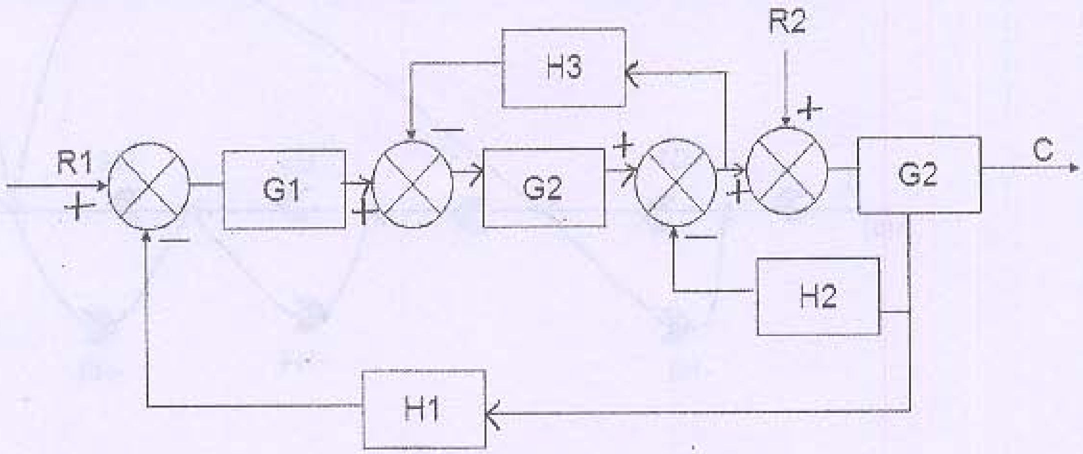


(b) A system has $G(s)H(s) = \dfrac{Ks^2}{(1+0.2s)(1+0.02s)}$.find the value of K so that gain crossover frequency is 5 rad/sec. Use bode plot. [07]

SECTION – II

Q - 4 For a system $G(s)H(s) = \dfrac{K}{s^2(s+2)(s+3)}$.Find the value of K to limit steady state error to 10 when input to the system is $1+10t+\dfrac{40t^2}{2}$. [07]

Q - 5 (a) For the system shown in figure determine C/R1 and C/R2. Assume R2=0 when r1 is applied and R1=0 when R2 is applied. Use block diagram Reduction. **[10]**



(b) Distinguish between open and close loop control system. **[4]**

OR

Q - 5 (a) A second order system is given by $\frac{C(s)}{R(s)} = \frac{25}{s^2+6s+25}$ .Find its rise time, peak time, peak overshoot and setting time if subjected to unit step input. Also determine the expression for its output response. **[07]**

(b) A system is described by **[07]**

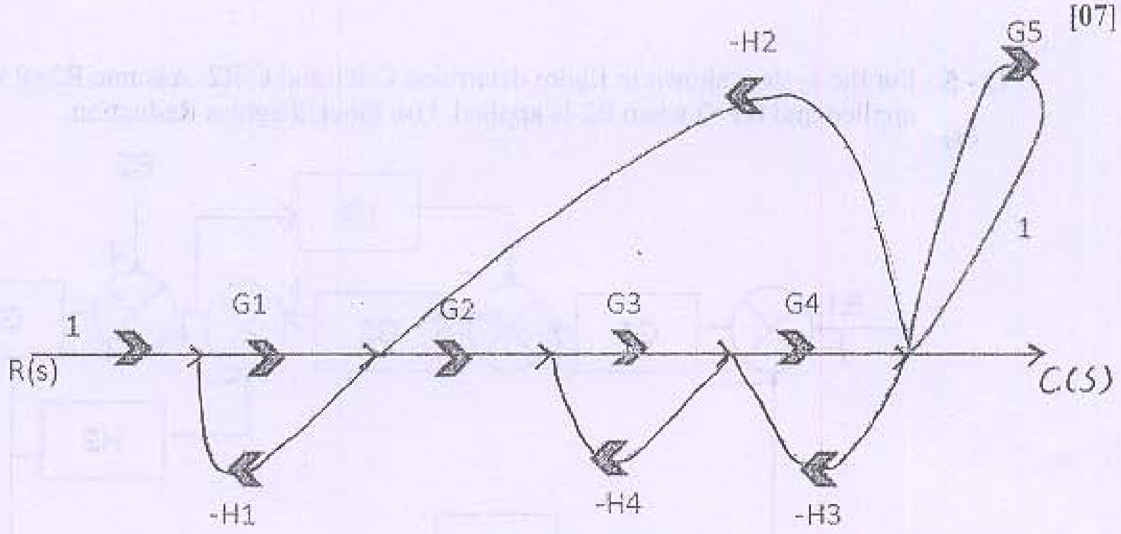
$$\begin{bmatrix} \dot{x}1 \\ \dot{x}2 \\ \dot{x}3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & -1 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} + \begin{bmatrix} 01 \\ 10 \\ 01 \end{bmatrix} u$$

$$Y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix}$$

Test its state controllability & observabillity.

Q - 6 (a) A feedback control system has open loop transfer function $G(s)H(s) = \frac{K(s+2)(s+3)}{(s-1)(s+1)}$ Find the root locus as K is varied from 0 to ∞ **[07]**

(b) [07]



Find $\dfrac{C(s)}{R(s)}$

OR

Q -6(a) Define Zero input Stability, Asymptotic Stability and Relative Stability [07]

(b) For the Gear Trains Derive, [07]

$$J_{1eq}\ddot{\theta}_1+f_{1eq}\dot{\theta}_1+\frac{N1}{N2}T_L=T_M$$

$$J_{2eq}\ddot{\theta}_2+f_{2eq}\dot{\theta}_2+T_L=\frac{N2}{N1}T_M$$

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